جامعة حفر الباطن
University of Hafr Al Batin

# Lecture 11 - Dynamic Host Configuration Protocol (DHCP)

# Objectives

- DHCP Overview.
- DHCO Operation.
- DHCP Error Control.
- DHCP Packet Format.
- DHCP Configuration.
- DHCP Transition States.

# DHCP Overview

- Each computer that uses the TCP/IP protocol suite needs <u>four pieces of information:</u>
  1. The IP address of the computer
  2. The subnet mask of the computer
  3. The IP address of a router (to communicate with other network)
  4. The IP address of a name server (to use names instead of addresses)
- These information can be stored in a configuration file and accessed by the computer.
- **But what about a diskless workstation or a computer with a disk that is booted for the first time?**
  - You can’t store this information in the ROM!

# DHCP Overview

- The **Dynamic Host Configuration Protocol (DHCP)** is a client/server protocol designed to provide the four pieces of information for a diskless computer or a computer that is booted for the first time.
- DHCP is a successor to Bootstrap Protocol (BOOTP) and is backward compatible with it.
  - Meaning that a host running the BOOTP client can request a static configuration to a DHCP server.

# DHCP Operation

- The DHCP client and server can either be on the same network or on different networks.

**1. Same Network:**

- Although the practice is not very common, the administrator may put the client and the server on the same network.

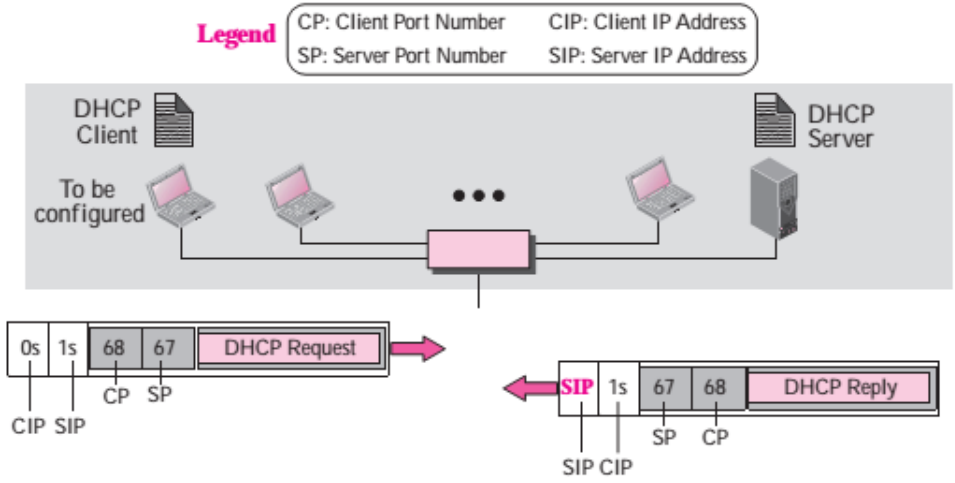
**1. Different Network:**

- As in other application-layer processes, a client can be in one network and the server in another, separated by several other networks.

# DHCP Operation

**Same Network Operation:**

1. The DHCP server issues a passive open command on UDP port number 67 and waits for a client.
2. A booted client issues an active open command on port number 68. The message is encapsulated in a UDP user datagram, using the destination port number 67 and the source port number 68. The UDP user datagram, in turn, is encapsulated in an IP datagram using all 0s as the source address and all 1s as the destination address.
3. The server responds with either a broadcast or a unicast message using UDP source port number 67 and destination port number 68.

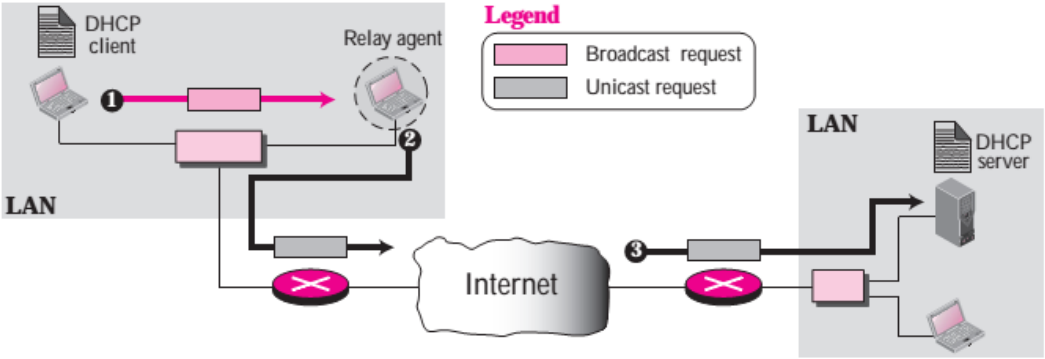


# DHCP Operation

## Different Network:

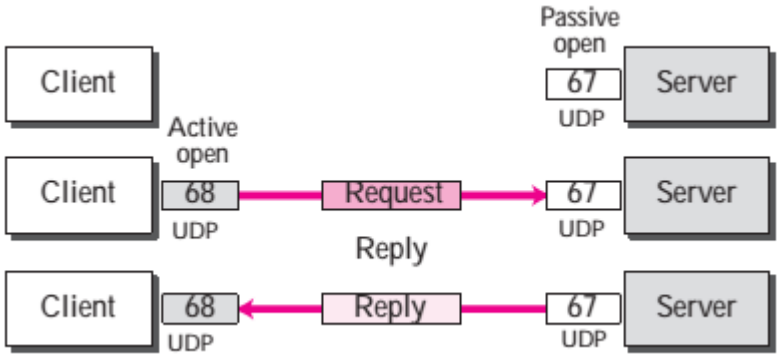
- There is one problem with this situation:
  - The DHCP request is broadcast.
    - Why? The client does not know the IP address of the server.
  - A broadcast IP datagram cannot pass through any router.
- Solution:
  - There is a need for an intermediary to be used as a relay.
    - The host in this case is called a **relay agent**.

1. The relay agent knows the unicast address of a DHCP server and listens for broadcast messages on port 67.
2. When it receives this type of packet, it encapsulates the message in a unicast datagram and sends the request to the DHCP server.
3. When it receives a response from the DHCP server, it will sends it to the DHCP client.



# DHCP & UDP Ports

- The server uses the well-known port 67.
- The client uses the well-known port 68.
  - The use of a well-known port (less than 1024) prevents the use of the same two destination port numbers.
  - Another host (in the same network) cannot select 68 as the ephemeral port.
- This figure shows the interaction between a client and a DHCP server:



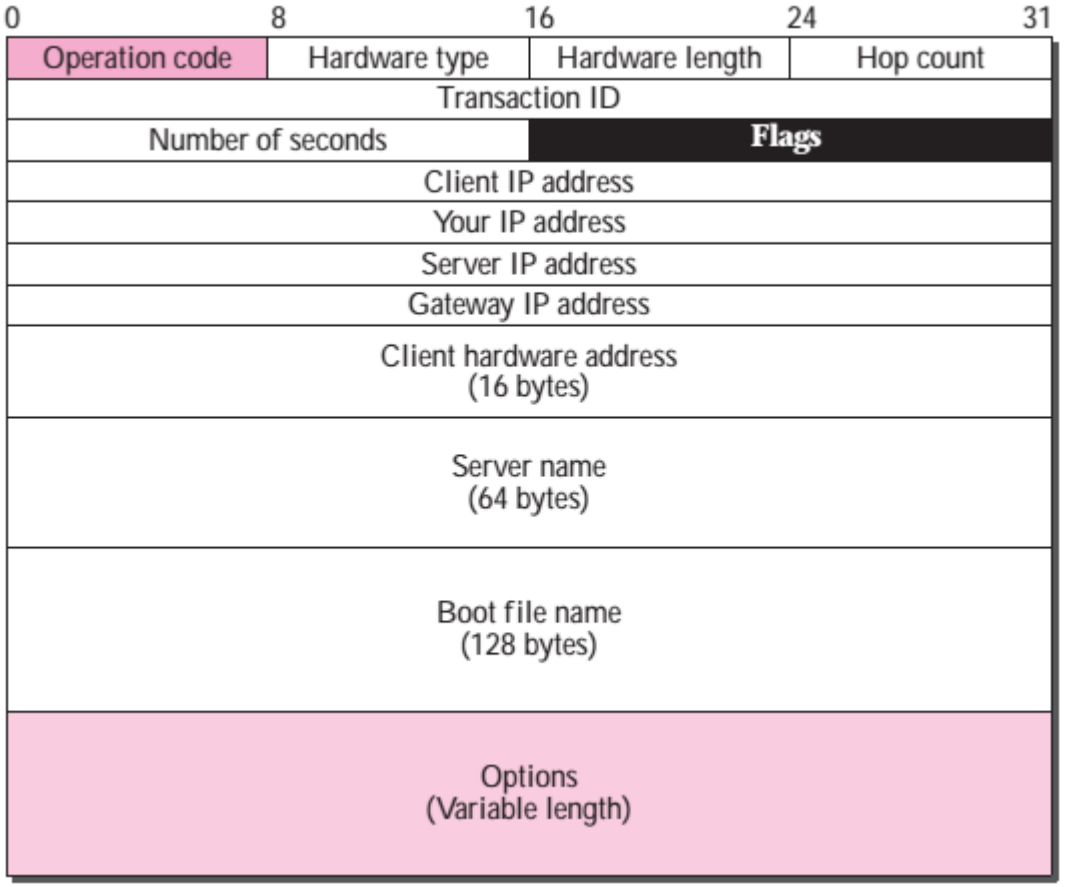
# DHCP Error Control

- **DHCP uses UDP, which does not provide error control.**
- Therefore, DHCP must provide error control.

- Error control in DHCP is accomplished through two strategies:
  1. **DHCP requires that UDP uses the checksum.**
     - Remember that the use of the checksum in UDP is optional.
  2. **The DHCP client uses timers and a retransmission policy if it does not receive the DHCP reply to a request.**
     - However, to prevent a traffic jam when several hosts need to retransmit a request (for example, after a power failure), DHCP forces the client to use a random number to set its timers..

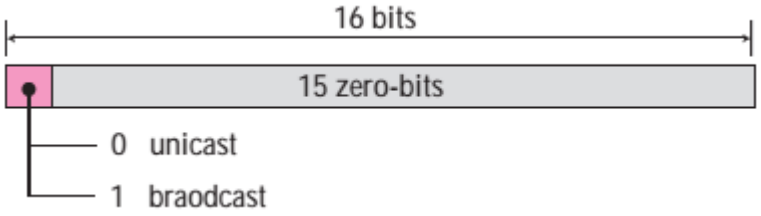
# DHCP Packet Format

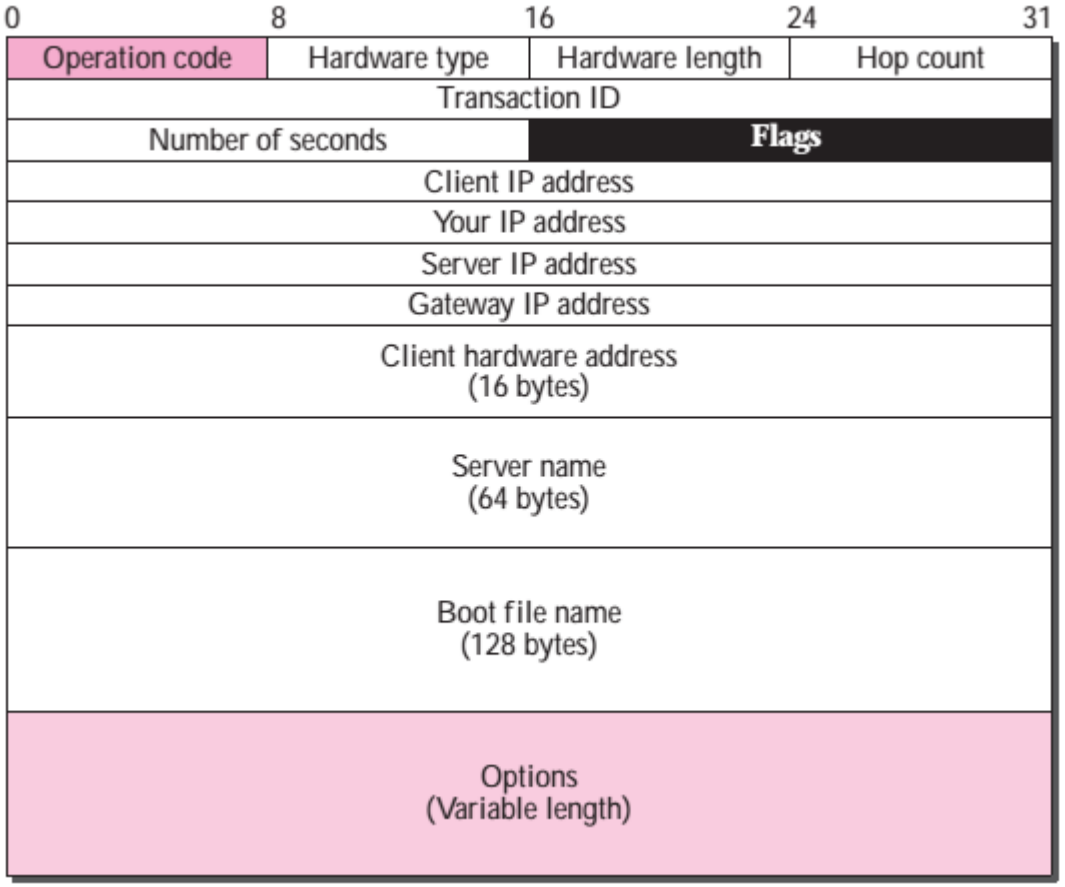
- **Operation code:** the type of DHCP packet
  - request (1) or reply (2).
- **Hardware type:** the type of physical network.
  - For example, for Ethernet the value is 1.
- **Hardware length:** the length of the physical address in bytes
  - For example, for Ethernet the value is 6.
- **Hop count:** the maximum number of hops the packet can travel.



# DHCP Packet Format

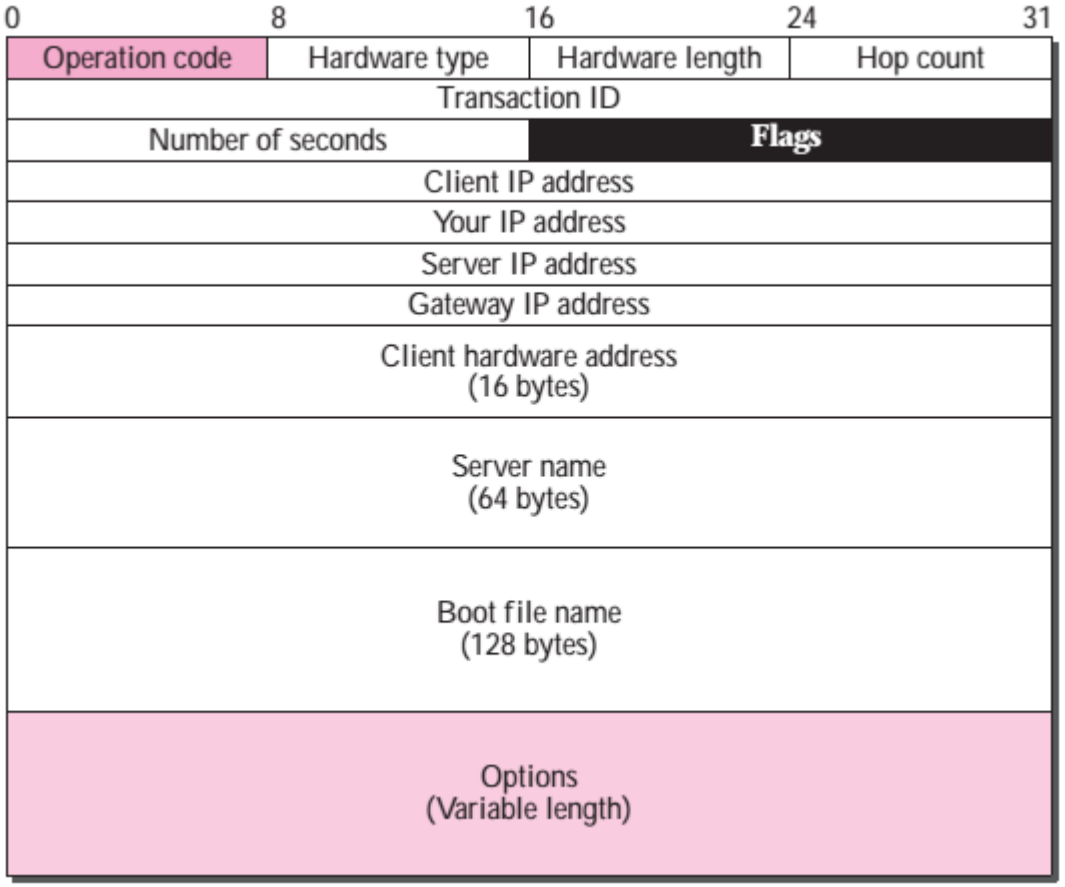
- **Transaction ID:** used to match a reply with the request.
  - The server returns the same value in its reply.
- **Number of seconds:** the number of seconds passes since the client booted.
- **Flags:** specifies a forced broadcast reply (instead of unicast) from the server
  - only the leftmost bit is used and the rest of the bits should be set to 0s.



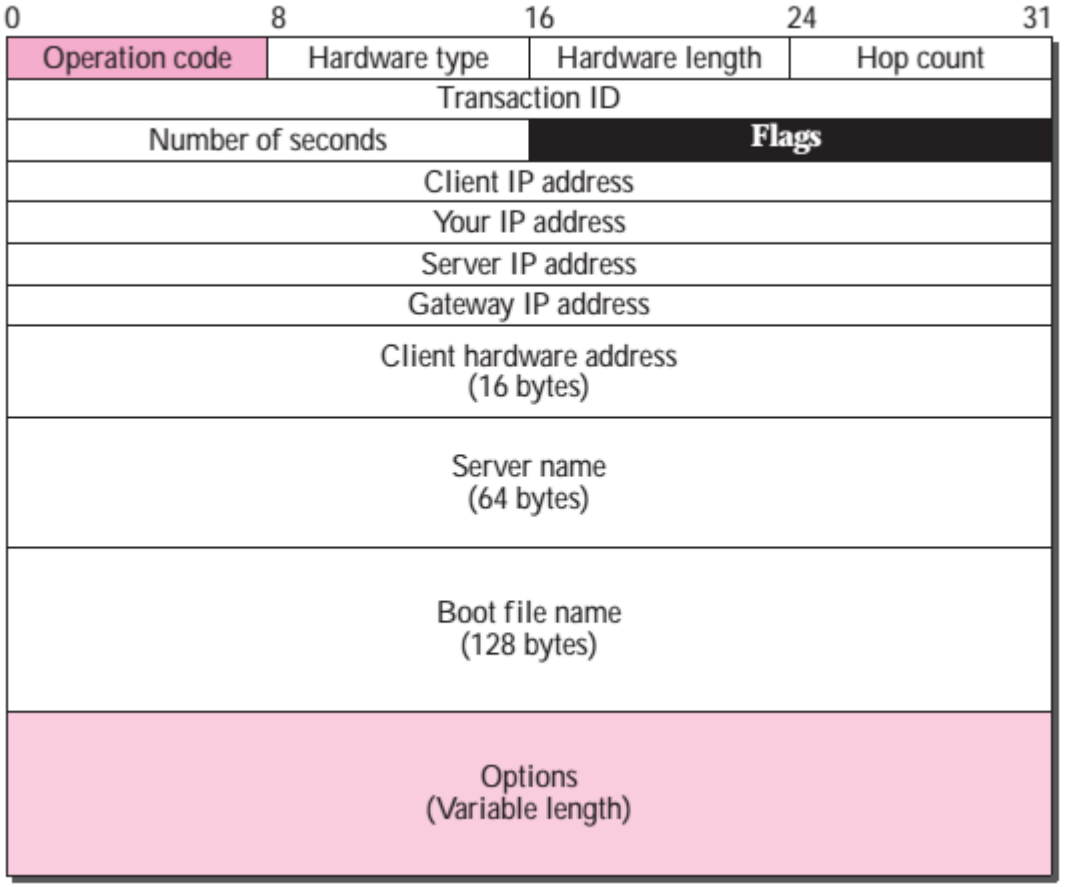
# DHCP Packet Format

- **Client IP:** If the client does not have this information, this field has a value of 0.
- **Your IP:** the client IP address.
  - filled by the server in the reply message.
- **Server IP:** the server IP address.
  - filled by the server in a reply message.
- **Gateway IP:** IP address of a router.
  - filled by the server in a reply message.
- **Client hardware address:** physical address of the client



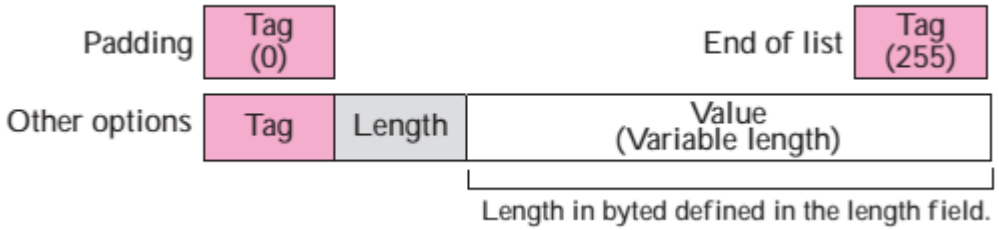
# DHCP Packet Format

- **Server Name:** the domain name of the server.
  - optionally filled by the server in a reply packet.
- **Boot filename:** full pathname of the boot file.
  - optionally filled by the server in a reply packet.
  - The client can use this path to retrieve other booting information.
- **Options:** It can carry either:
  1. additional information (such as the network mask or default router address)
  2. or some specific vendor information.
  - The field is used only in a reply message.



# DHCP Packet Format

- A DHCP packet option is composed of three fields:
  1. 1-byte tag field.
  2. 1-byte length field.
  3. Variable-length value field.
- The length field defines the length of the value field, not the whole option.



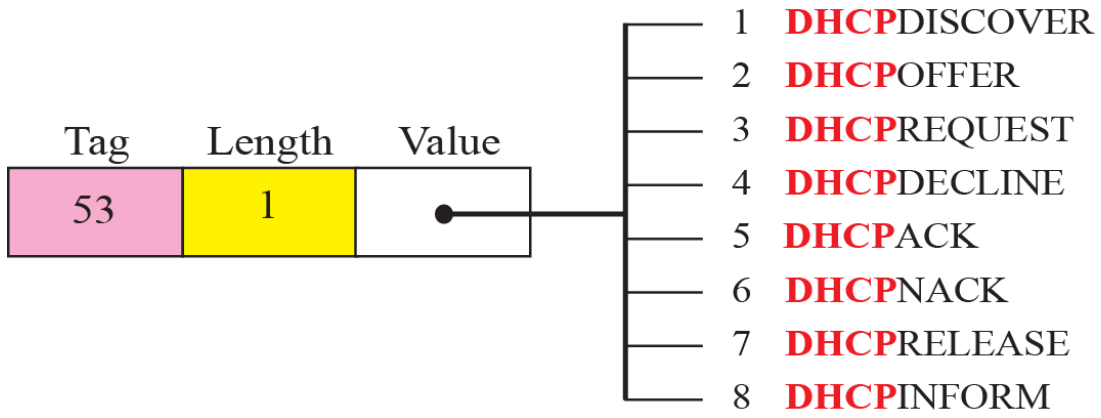

| *Tag* | *Length* | *Value* | *Description* |
|---|---|---|---|
| 0 | | | Padding |
| 1 | 4 | Subnet mask | Subnet mask |
| 2 | 4 | Time of the day | Time offset |
| 3 | Variable | IP addresses | Default router |
| 4 | Variable | IP addresses | Time server |
| 5 | Variable | IP addresses | IEN 16 server |
| 6 | Variable | IP addresses | DNS server |
| 7 | Variable | IP addresses | Log server |
| 8 | Variable | IP addresses | Quote server |
| 9 | Variable | IP addresses | Print server |
| 10 | Variable | IP addresses | Impress |
| 11 | Variable | IP addresses | RLP server |
| 12 | Variable | DNS name | Host name |
| 13 | 2 | Integer | Boot file size |
| 53 | 1 | Discussed later | Used for dynamic configuration |
| 128–254 | Variable | Specific information | Vendor specific |
| 255 | | | End of list |

# DHCP Configuration

- The DHCP has been devised to provide static and dynamic address allocation.
- Static Address Allocation:
  - In this capacity, a DHCP server has a database that statically binds physical addresses to IP addresses.
  - When working in this way, DHCP is backward compatible with the deprecated protocol BOOTP.
- Dynamic Address Allocation:
  - DHCP has a second database with a pool of available IP addresses.
  - DHCP will check its static database before assigning an IP from the pool.
  - When a DHCP client requests a temporary IP address, the DHCP server goes to the pool of available (unused) IP addresses and assigns an IP address for a negotiable period of time.
  - DHCP server issues a **lease** for a specific period of time. When the lease expires, the client must either stop using the IP address or renew the lease.
    - The server can agree or disagree with the renewal.

# DHCP Transition States

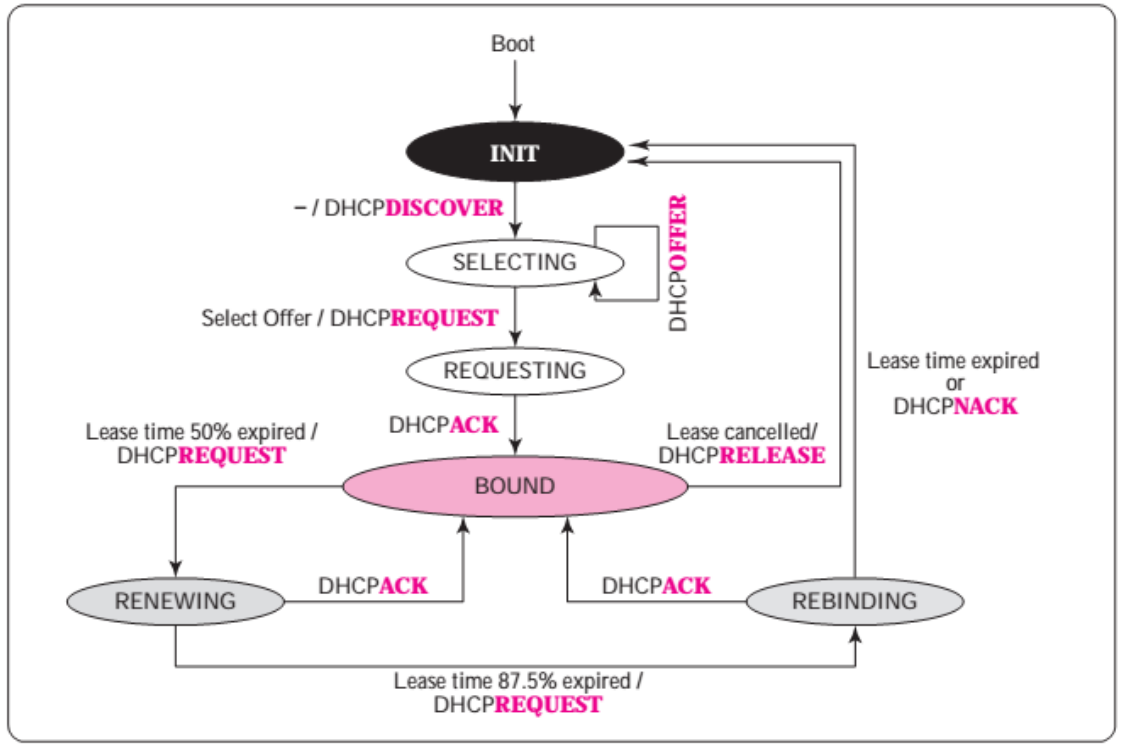
- DHCP client acts as a state machine that performs transitions from one state to another depending on the messages it receives or sends.
- The type of the message in this case is defined by the option with tag 53 that is included in the DHCP packet.
- Instead of adding one extra field to the BOOTP protocol to define DHCP type, the designer decided to add an extra option for this purpose.



# DHCP Transition States

**INIT State:**

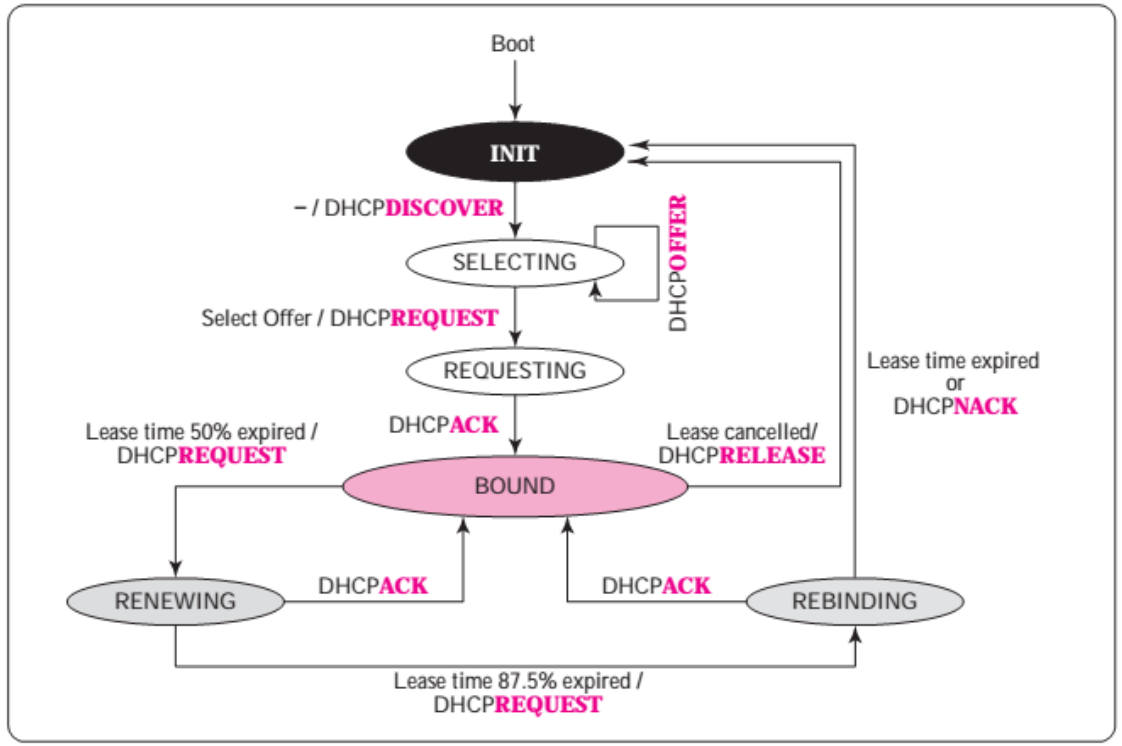
- When the DHCP client first starts, it is in the INIT state (initializing state).
- The client broadcasts a DHCPDISCOVER message using port 67.



# DHCP Transition States

## SELECTING State:

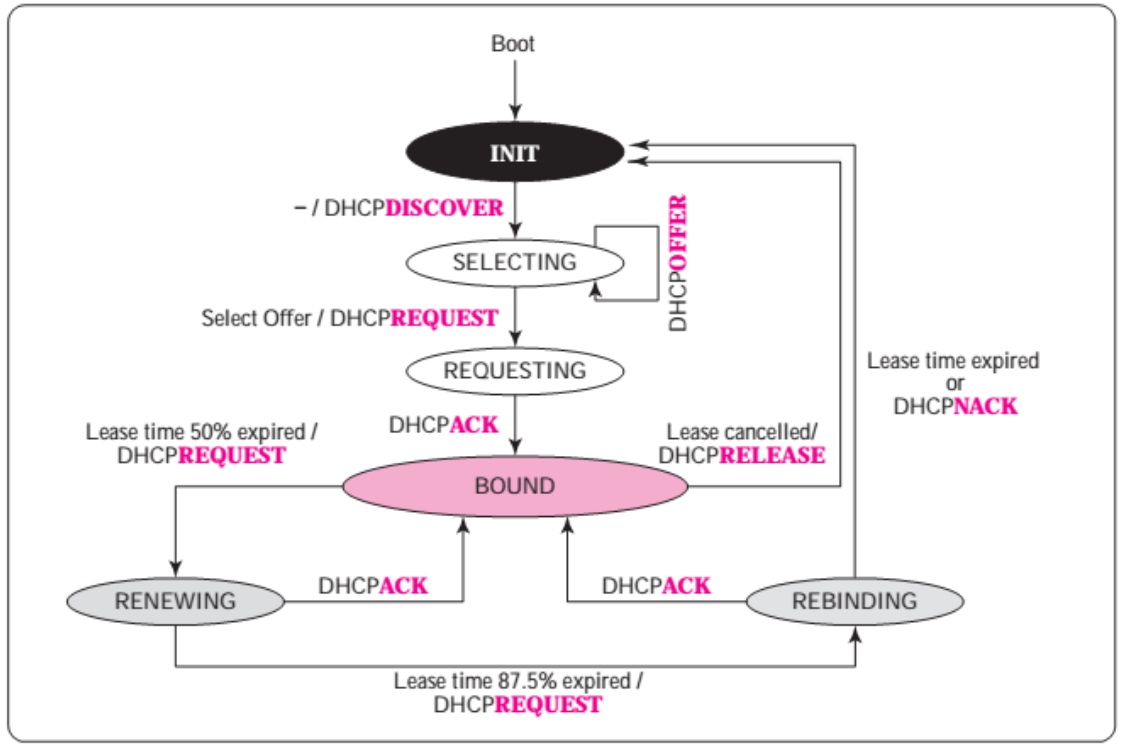
- Those servers that can provide this type of service respond with a DHCPOFFER message.
  - In these messages, the servers offer an IP address.
  - They can also offer the lease duration. The default is 1 hour.
  - The server that sends a DHCPOFFER locks the offered IP address so that it is not available to any other clients.
- The client choose one offers and sends a DHCPREQUEST message to the selected server.



# DHCP Transition States

## REQUESTING State:

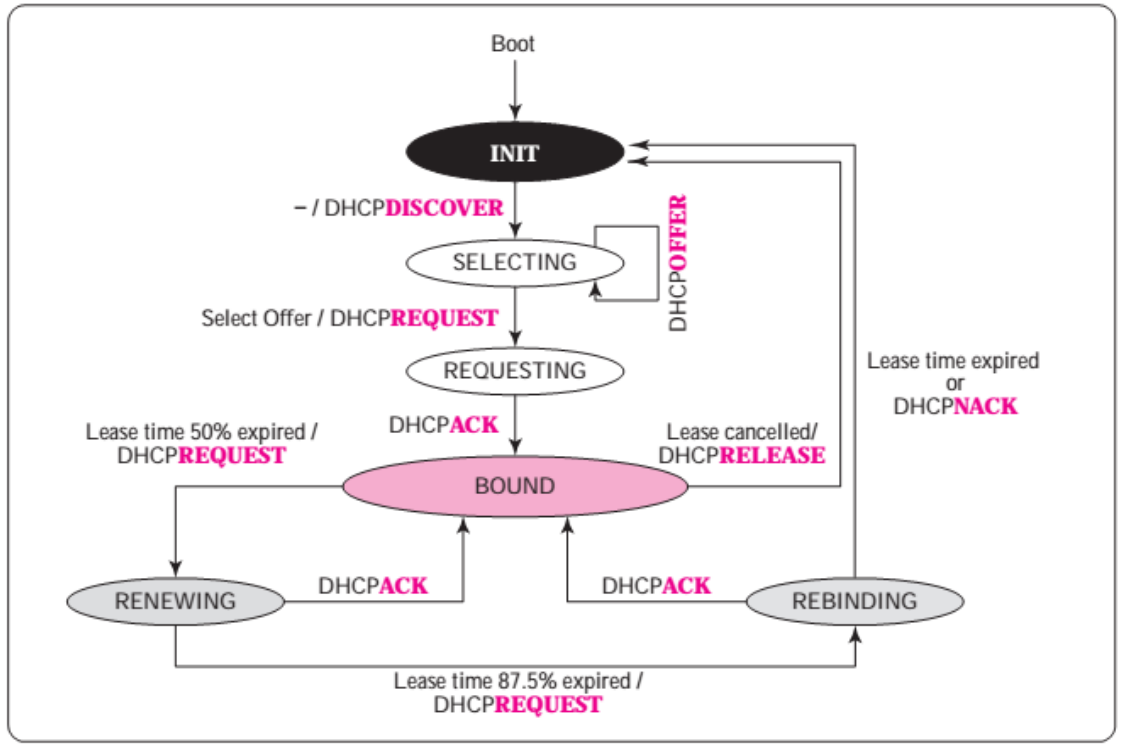
- The client remains in the requesting state until it receives a DHCPACK message from the serve.
- After receipt of the DHCPACK, the client goes to the bound state.



# DHCP Transition States

**BOUND State:**

- In this state, the client can use the IP address until the lease expires.
- When 50 percent of the lease period is reached, the client sends another DHCPREQUEST to ask for renewal.
  - It then goes to the renewing state.
- When in the bound state, the client can also cancel the lease.
  - … and go to the initializing state.

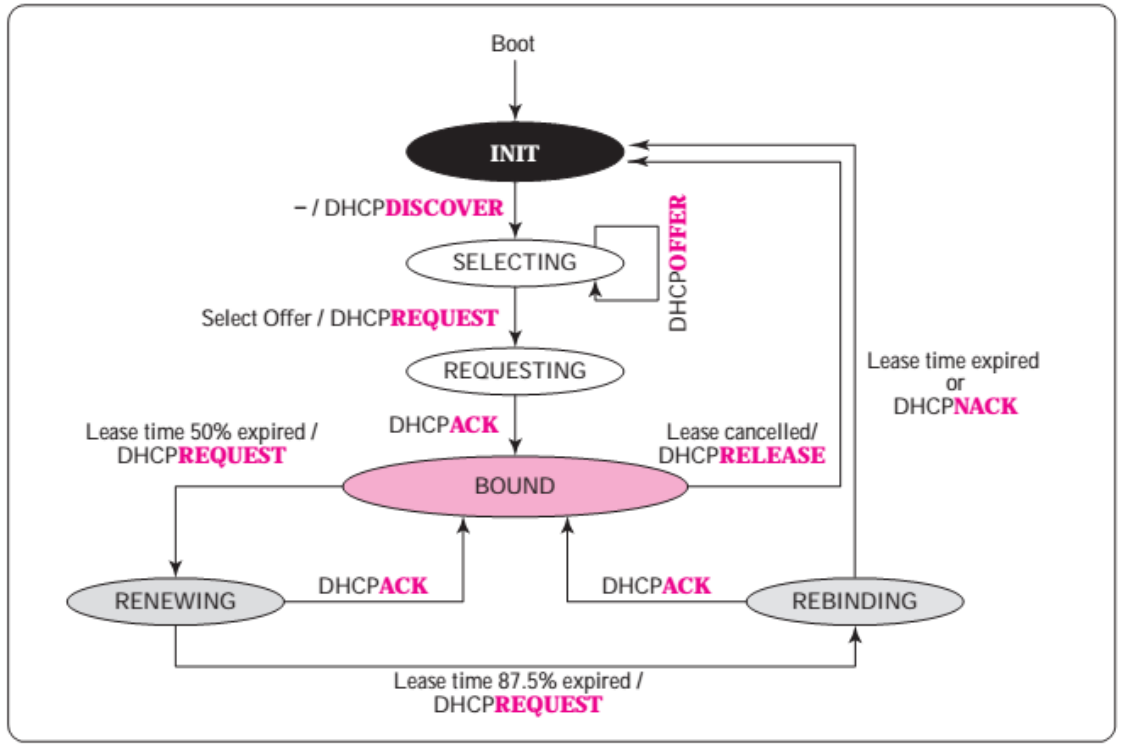


# DHCP Transition States

## RENEWING State:

- The client remains in the renewing state until one of two events happens.

1. A DHCPACK is received, which renews the lease agreement:
   - In this case, the client resets its timer.
   - and goes back to the bound state.
2. A DHCPACK is not received, and 87.5 percent of the lease time expires:
   - the client goes to the rebinding state.

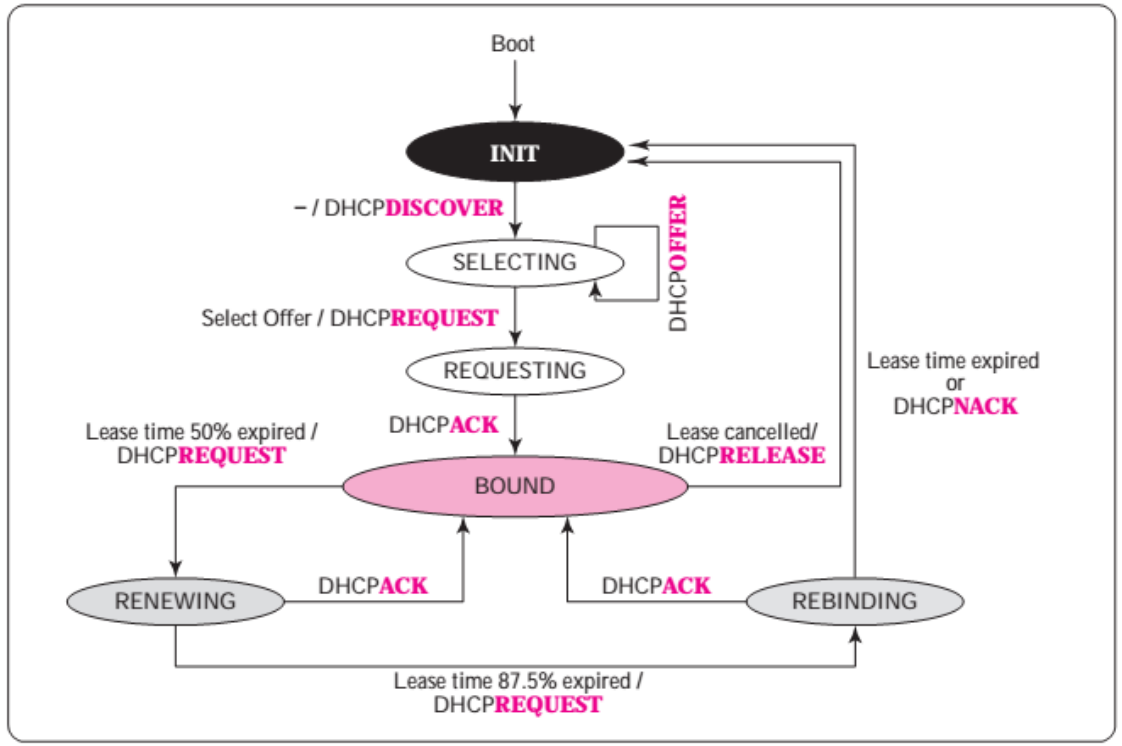


# DHCP Transition States

**REBINDING State:**

- The client remains in the rebinding state until one of three events happens.

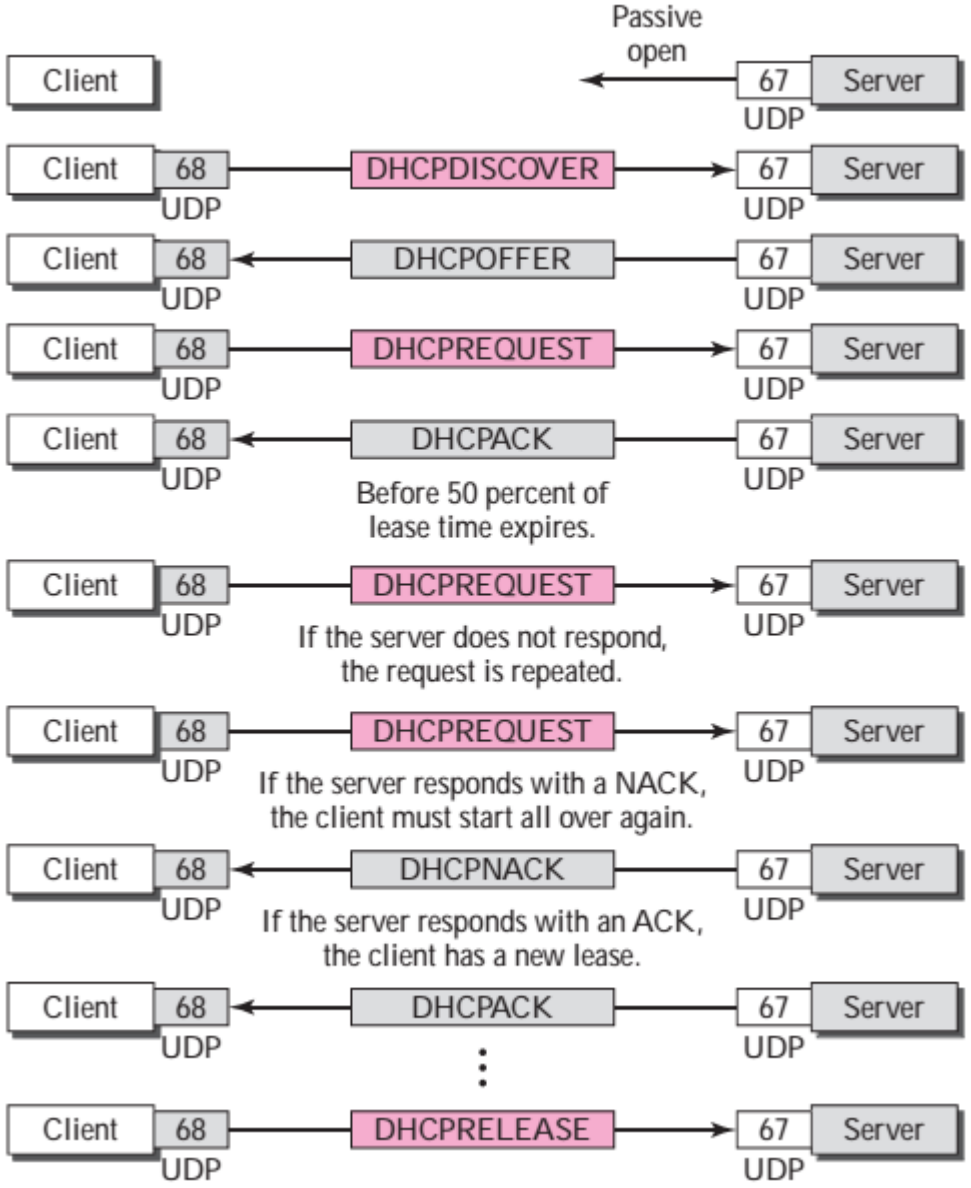
1. A DHCPNACK is received:
   - it goes back to the initializing state and tries to get another IP address.
2. the lease expires:
   - it goes back to the initializing state and tries to get another IP address.
3. A DHCPACK is received:
   - it goes to the bound state and resets the timer.



# DHCP Timers

- The above discussion requires that the client uses three times: renewal timer, rebinding timer, and expiration timer.
- If the server does not specify the time-out values for these timers when the address is allocated, the client needs to use the default value.
- The default value for each timer is shown below:

| | | |
|---|---|---|
| **Renewal timer:** | → | **50% of lease time** |
| **Rebinding timer:** | → | **87.5% of lease time** |
| **Expiration timer:** | → | **100% of lease time** |

# DHCP Exchanging Messages

- This figure shows the exchange of messages related to the transition diagram.